Ariel Gjaci

Postdoctoral Researcher | Multimodal Generative Learning | Spatial Audio | Human Motion

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Website | Linkedin | GitHub | Google Scholar

EDUCATION

Ph.D. Artificial Intelligence & Robotics

University of Genoa

Nov 2021 - May 2025 (defended)

Research collaboration with King's College London

M.Sc. Robotics Engineering

University of Genoa

2018 - 2021 | 103/110

Thesis: Culture-Aware Co-Speech Gestures using GANs

B.Sc. Biomedical Engineering

University of Genoa

2014 - 2018 | 92/110

Thesis: Activity in Hippocampal Neuronal Networks

CORE SKILLS

PROGRAMMING

Python; C/C++; R; MATLAB/Simulink; LaTeX

ML FRAMEWORKS

PyTorch; Scikit-learn; TensorFlow

METHODS

Diffusion transformers; flow matching; neural audio codecs; domain generalization

RESEARCH SYSTEMS

Linux; SLURM; multi-GPU training; Git; ROS 1

LANGUAGES

Italian (native)
English (fluent)
Albanian (native)

TRAINING

Modern Topics in Machine Learning (MaLGa); Robotics Summer School (Scuola Superiore Sant'Anna); Coursera DL/NLP courses.

PROFESSIONAL EXPERIENCE

ITALIAN INSTITUTE OF TECHNOLOGY | Postdoctoral Researcher

Feb 2025 - Present | Genoa, Italy

- Developing FACT-RIR, a unified pipeline combining neural audio codecs and conditional flow matching for few-shot room impulse response generation from sparse acoustics, geometry, and visual evidence; investigating room-level and query-dependent representations and physics-informed conditioning.
- Built 3D sound-source localization on room point clouds using MVDR, DAS, and MaxRE beamforming.
- Curating multimodal spatial-audio data and writing a survey on learning-based spatial audio rendering.

UNIVERSITY OF GENOA & KCL | PhD Researcher

Nov 2021 - May 2025

- Developed rule-based and GAN-based culture-aware co-speech gestures for social robots (AIRO; International Journal of Social Robotics).
- Analyzed cultural cues in multimodal interactions with classical and deep learning models (AIRO; AHRI @ ACII).
- Developed semantic-similarity algorithms for symbolic and deictic gesture labeling (IEEE RO-MAN 2024).
- Built TED4C-L: 106.45 hours from 764 speakers across four cultures; learned speaker-independent culture embeddings with Fishr and adversarial learning.
- Designed ALaDiT, a diffusion-transformer for culture-aware motion generation, as part of SICAGE (ECCV 2026, main conference).

AKKA TECHNOLOGIES | Software Engineer

May 2021 - Oct 2021 | Italy

- Developed Python backend software for aircraft panels and presented a working demo to stakeholders.

TEACHING & SERVICE

- Teaching assistant: Foundations of Informatics (C++), 60h; Mobile Programming (Android/Kotlin), 30h, University of Genoa.
- Co-organizer, ECCV workshop Expressive Encounters; session chair, IEEE RO-MAN 2024.
- Reviewer: AAAI, ECCV, ACM Multimedia, IROS, ICMI, and International Journal of Social Robotics.

PUBLICATIONS

Gjaci A., Sgorbissa A., Murino V. SICAGE: Speaker-Independent Culture-Aware Gesture Generation using TED4C-L Dataset. ECCV 2026, main conference, pp. 392-411.

Gjaci A., Schmuck V., Sgorbissa A., Celiktutan O. Exploring Cultural Cues in Multimodal Interactions. AHRI @ ACII Workshop, 2024.

Gjaci A., Recchiuto C.T., Sgorbissa A. Labeling Sentences with Symbolic and Deictic Gestures via Semantic Similarity. IEEE RO-MAN, 2024.

Gjaci A., Recchiuto C.T., Sgorbissa A. Towards Culture-Aware Co-Speech Gestures for Social Robots. Int. Journal of Social Robotics, 2022.

Gjaci A., Oneto L., Recchiuto C.T., Sgorbissa A. Culture Awareness in Intelligent Systems. AIRO Workshop, 2022.

Gjaci A., Recchiuto C.T., Sgorbissa A. A GAN-based Approach for Generating Culture-Aware Co-Speech Gestures. AIRO Workshop, 2021.